# ■# PAPER<i>197: F</i>U<i>Bi</i>i Extended Integral — UV, mm-Wave, Hybrid, and Hierarchical Terms

**Author:** Star-Magic UQFF Research Framework

*Session: 50 — grokshare7514fe.txt Full Audit*

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■# PAPER*197: FUBi*i Extended Integral — UV, mm-Wave, Hybrid, and Hierarchical Terms

**Version:** 1.0 **Date:** March 13, 2026 **Session:** 50 — grok*share*7514fe.txt Full Audit **Author:** Star-Magic UQFF Research Framework **Source:** grok*share*7514fe.txt lines 200–400

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```
F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57
```

```
U_{b_i}(r) = \kappa\cdot[SSq]\cdot\frac{GM}{r^2}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57,\; \beta_i = 0.61
```

<!-- ? = 5.0e-4 day?[1], [SSq] = 0.57, ß_i = 6.1e-1 -->

## Abstract

This paper documents the extended form of the F*U*Bi*i buoyancy integral, incorporating four new terms beyond the standard UQFF formulation: F*UV (GALEX/Spitzer UV flare coupling), F*mm (ALMA mm-radio coupling), F*hyb (hybrid polarization-frequency term), and F*hier (hierarchical remnant unification). Numerical coefficients k*UV = kmm = *10?$^{3}$∘ N/W and f*mm = 1.05 are derived from observational calibration. This extended integral enables multi-wavelength coupling of buoyancy forces from UV through millimeter radiation fields.

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## 1. Standard F*U*Bi_i Integral (Baseline)

The standard buoyancy integral form:

```
F_U_Bi = -F0 + (m_e c²/r²)·DPM_momentum·cos? + (GM/r²)·DPM_gravity + F_U_Bi_i
```

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## 2. Extended F*U*Bi_i Integral

The complete extended form including all observational coupling terms:

```
F_U_Bi_i = ?0^{x2} [
-F0
+ (m_e c²/r²) · DPM_mom · cos?
+ (GM/r²) · DPM_grav
+ ?_vac,[UA] · DPM_stab
+ k_LENR · (?_LENR/?0)²
+ k_act · cos(?_act · t)
+ k_DE · L_X
+ 2qB0V · sin? · DPM_res · P_pol
+ k_neutron · s_n
+ k_rel · (E_cm,astro,enhanced/E_cm)²
+ k_UV · L_UV ? NEW: UV flare coupling
```

```
+ k_mm · L_mm · f_mm ? NEW: mm-radio coupling
] dx
```

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## 3. New Extended Terms

### *3.1 F_UV — GALEX/Spitzer UV Coupling*

```
F_UV = k_UV · L_UV
Parameters:
k_UV = 10?³° N/W (UV force coupling constant)
L_UV = UV luminosity (W) from GALEX or Spitzer photometry
Physical origin: UV flare irradiation pressure on buoyant plasma shells
```

### *3.2 F_mm — ALMA mm-Radio Coupling*

```
F_mm = k_mm · L_mm · f_mm
Parameters:
k_mm = 10?³° N/W (mm-radio force coupling constant)
L_mm = mm-radio luminosity (W) from ALMA observations
f_mm = 1.05 (mm-radio frequency enhancement factor)
Physical origin: millimeter-wave radiation pressure in molecular cloud outflows
```

### *3.3 F_hyb — Hybrid Polarization-Frequency Term*

```
F_hyb = P_pol · f_mm · (?0)?¹
Parameters:
P_pol = polarization fraction (dimensionless, typically 0.01–0.1)
f_mm = 1.05
?0 = base UQFF angular frequency (rad/s)
Physical origin: coupling of polarized mm-emission to buoyancy oscillation frequency
```

### *3.4 F_hier — Hierarchical Remnant Unification*

```
F_hier = S? (v_i/c)^n · ?0^{-m}
Parameters:
v_i = velocity of remnant component i (m/s)
c = speed of light
n = 2 (hierarchical power index)
m = 1 (frequency suppression exponent)
Physical origin: multi-component remnant velocity hierarchy (e.g., jet + cocoon + lobe)
```

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## 4. Standard Integral Terms (Reference)

| Term | Symbol | Physical Origin |
|---|---|---|
| Base restoring force | -F0 | Vacuum restoring force |
| DPM momentum | (me $c^2/r^2$)·*DPM*mom·cos? | Dipole-plasma momentum scattering |
| DPM gravity | (GM/$r^2$)·DPM_grav | Dipole-plasma gravitational coupling |
| DPM stability | ?*vac,[UA]·DPM*stab | Vacuum aether stability term |
| LENR coupling | k*LENR*·(?LENR/?0)$^2$ | Low-energy nuclear resonance |
| Activation term | k*act·cos(?*act·t) | Activation oscillation |
| Dark energy luminosity | k*DE·LX* | X-ray dark energy coupling |

| Term | Symbol | Physical Origin |
|---|---|---|
| DPM resonance | 2qB0V·sin?·DPM*res*·*P*pol | Magnetic resonance polarization |
| Neutron cross-section | *kneutron*·sn | Neutron scattering |
| Relativistic CM | k*rel*·(*E*cm,astro,enh/E_cm)$^2$ | Relativistic center-of-mass enhancement |

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## 5. Numerical Results for Extended Terms

| System | Standard *FU*Bi_i | With UV/mm extension |
|---|---|---|
| Magnetar SGR1745 | ˜ 2.11×10$^2$°8 N | + F_UV from Chandra/GALEX |
| NGC 3603 | ˜ -8.31×10$^{211}$ N | + F_mm from ALMA CO observations |
| Pillars of Creation | ˜ 9.79×10?$^{33}$ N | + F*hyb with P*pol ˜ 0.05 |

**Note:** The extreme magnitudes (10$^2$°8 N, 10$^{211}$ N) reflect vacuum-density-scaled units in the UQFF framework where ?_vac,[UA] ˜ 10?$^{113}$ (dimensionless normalized).

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## 6. Integration with Existing UQFF Terms

The extended integral fits within the Triadic Master System (PAPER*196) as the primary buoyancy channel FU*Bi. Specifically:

- **F_UV** activates when GALEX UV flux > threshold (flare events)
- **F_mm** activates when ALMA observes SFR-driven mm continuum
- **F_hyb** operates continuously as polarization coupling
- **F_hier** applies to multi-component systems (AGN jets, SNR shells)

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## 7. References


- `grok_share_7514fe.txt` lines 200–400 (first PDF extended integral section)
- PAPER_196: Triadic Master Equation System
- PAPER_171: Universal Gravity Ug1–Ug4 Decomposition
- PAPER_172: FU Complete Unified Field Assembly


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